

Course: B. Tech.

Branch: Civil Engineering

Semester: V

Subject Code & Name: BTHM505 Project Management

Max Marks: 60

Date: 19/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

- A) Discuss Significance of Project Management. 6
- B) Explain Bar chart and Milestone chart in detail. 6
- C) Determine the Project Duration (Days) and Critical Path for the following: 6

Activity	A	B	C	D	E	F	G	H
Predecessor	-	A	A	B	C	C	D, E	F
Duration	5	7	8	8	10	12	8	8

Q.2 Solve Any Two of the following.

12

- A) Discuss the concept of cost slope and network compression in detail. 6
- B) Find the Project Duration (Days) and Project Cost (Rs.). 6

Activity	Normal Time	Crash Time	Normal Cost	Crash Cost
1-2	6	5	6000	9000
1-3	7	4	5000	15000
2-4	6	4	10000	16000
2-5	7	7	3000	3000
3-5	5	4	7000	8500
4-6	9	7	4000	12000
5-6	7	4	5000	23000

- C) Crash the network given in Q. 2 (B)

6

Q. 3 Solve Any Two of the following.

12

- A) Discuss PERT in detail. 6
- B) Draw the network and Determine the Project Duration. 6

Activity	Optimistic Time	Most likely time	Pessimistic time

1-2	6	10	20
1-3	7	12	17
2-4	9	11	13
2-5	7	10	13
3-5	3	7	17
4-6	9	9	9
5-6	7	10	13

- C) Find the Z value if the Project mentioned in Q.3 B. has to be completed in 25 days.

6

Q.4 Solve Any Two of the following.

12

- A) Explain Demand and Supply curve, Inflation and Cash flow diagram.
- B) A company bought a Crane at Initial Investment of 50 lakhs. The Annual Operation and Maintenance charges are 2 Lakhs. The company earns 10 lakhs in Annual Revenue. The Salvage value of the Equipment is 5 lakhs after 10 years of service. Draw the Cash flow diagram for above transaction and find the expenses and earning.
- C) Explain various interest formulae.

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Q. 5 Solve Any Two of the following.

12

- A) Fixed Cost associated with a product is 70 Lakhs, variable cost is Rs. 25 and the revenue from each item is Rs. 80. Calculate the Break-even quantity if the targeted profit is Rs. 10 Lakhs.
- B) Discuss the significance of quality control in Project Management.
- C) Explain ISO 14000 in detail.

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***** End *****